

Maths 141H – Homework 4

Due: **Beginning of class on 12/2/2024**

Rules

- This homework consists of **5 questions**, all carrying equal weight (although they are of unequal difficulty!). Please attempt all the questions because credit will be given for all well-written and neat attempts which show effort! Your grade will be determined by **the four questions on which you have scored the highest**. The fifth question will be extra credit (up to 12.5%).
- Work in groups of **2-3 people**. Each group should submit **a single solution**, clearly indicating the group members (names and UIDs are required). **Groups must have been decided by 11/25** and announced to me by email, before the end of the working day. If you cannot find a group, let me know. **All members of the same group will receive the same grade.**
- Submit your answers at the beginning of class on 12/02/2024. Ensure your answers are legible (illegible parts will be ignored), and to help with the **markinggrading**,¹ please answer each question on a fresh page and staple everything together.
- **Good luck!**

¹A happy **markergrader** is a generous marker! Keep this in mind.

Question 1 – Geometric Series

A **geometric series** is a series of the form: $\sum_{n=m}^{\infty} cr^n$, for constants r, c , with $c \neq 0$. We call r the **common ratio** of the series. In class we proved that $\sum_{n=1}^{\infty} r^n$ converges if, and only if, $|r| < 1$ and, that in this case:

$$\sum_{n=1}^{\infty} r^n = \frac{r}{1-r}.$$

- (1) Prove, more generally, that if $|r| < 1$, $m \in \mathbb{N}$ and $c \in \mathbb{R} \setminus \{0\}$, then:

$$\sum_{n=m}^{\infty} cr^n = \frac{cr^m}{1-r}.$$

An interesting relationship exists between infinite geometric series and **repeating decimal expansions**. Let $k \in \mathbb{N}_{\geq 1}$ and $d_1, \dots, d_k$ be natural numbers between 0 and 9. We write:

$$0.\overline{d_1 \cdots d_k} = 0.(d_1 \cdots d_k)(d_1 \cdots d_k)(d_1 \cdots d_k) \dots,$$

so, for example, $0.\overline{3} = 0.333\dots$, $0.\overline{12} = 0.121212\dots$, etc.

- (2) Let $k \in \mathbb{N}_{\geq 1}$ and $d_1, \dots, d_k$ be natural numbers between 0 and 9. Express $0.\overline{d_1 \cdots d_k}$ as the sum of a geometric series.
- (3) Apply your answer from (2) to express $0.\overline{3}$ and $0.\overline{12}$ as infinite series. Then, using (1), or otherwise, find integers $a, b, c, d \in \mathbb{N}_{\geq 1}$ such that $\frac{a}{b} = 0.\overline{3}$ and $\frac{c}{d} = 0.\overline{12}$.
- (4) Generalise your answer from (3) to arbitrary repeating decimal expansions. More precisely: given $k \in \mathbb{N}_{\geq 1}$ and natural numbers $d_1, \dots, d_k$ between 0 and 9 find natural numbers a, b such that $\frac{a}{b} = 0.\overline{d_1 \cdots d_k}$ as a fraction.

In class, I joked about $\sum_{n=1}^{\infty} n = -\frac{1}{12}$ (if you're interested, look it up on Wikipedia). Here's a more absurd example:

- (5) Consider the following argument: Let $a = \sum_{n=1}^{\infty} 2^n = 1 + 2 + 4 + 8 + \dots$. Clearly: $2a = \sum_{n=1}^{\infty} 2^{n+1} = 2 + 4 + 8 + \dots = a - 1$. Thus, $2a = a - 1$, so $a = -1$. We just proved that:

$$1 + 2 + 4 + 8 + \dots = -1.$$

This is utterly absurd! What is the mistake in the previous argument?

Question 2 – There is no “smallest” divergent series

The goal of this question is to show that if $\sum_{n=1}^{\infty} a_n$ is a positive divergent series then there is a divergent series $\sum_{n=1}^{\infty} b_n$ such that for all $n \geq 1$ we have that $0 < b_n < a_n$. The way to understand this is the following: there is *no* “smallest” positive divergent series.

(1) Show that $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$ diverges.

(3) Show that $\sum_{n=3}^{\infty} \frac{1}{n \ln(n) \ln(\ln(n))}$ diverges.

(3) Show that:

$$\frac{1}{n} \geq \frac{1}{n \ln(n)} \geq \frac{1}{n \ln(n) \ln(\ln(n))},$$

for all $n \geq 16$. So the series from (2) diverges “more slowly” than the series from (1). Then, without using the next parts of this question, find a series that diverges even more slowly than the series in (2).

(4) Suppose that $\sum_{n=1}^{\infty} a_n$ is a series with positive terms which diverges. Let

$$b_n = \frac{a_n}{a_n + 1}$$

Show that $\sum_{n=1}^{\infty} b_n$ diverges and that $0 \leq b_n \leq a_n$, for all $n \in \mathbb{N}_{\geq 1}$. Using these ideas, find a series that is different from the one you found in Part 3 and that diverges more slowly than the series in Part 2.

The next part of the question is only tangentially related to the previous parts. It is something of an analogue of the p -series test, but for series involving logarithms:

(5) Show that $\sum_{n=2}^{\infty} \frac{1}{(\ln(n))^n}$ converges, but for any fixed $k \in \mathbb{N}_{\geq 1}$ the series $\sum_{n=2}^{\infty} \frac{1}{(\ln(n))^k}$ diverges

Question 3 – The Ratio and Root Tests

In the lectures, I claimed that the Root Test is “*better*” than the Ratio Test. The first aim of this question is to (formally) justify this statement.

- (1) Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of positive numbers. Show that:
 - (a) If $\sum_{n=1}^{\infty} a_n$ converges by the Ratio Test, then it converges by the Root Test.
 - (b) If $\sum_{n=1}^{\infty} a_n$ diverges by the Ratio Test, then it diverges by the Root Test.
 - (c) Suppose that the Root Test is inconclusive for $\sum_{n=1}^{\infty} a_n$, that is suppose that $\lim_{n \rightarrow \infty} \sqrt[n]{a_n}$ exists. Show that if $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$ then $L = 1$, that is, the Ratio Test is also inconclusive for $\sum_{n=1}^{\infty} a_n$.
- (2) Let $(b_n)_{n \in \mathbb{N}}$ be the sequence defined as follows:

$$b_n = \begin{cases} \frac{1}{n^n} & \text{if } n \text{ is odd;} \\ \frac{1}{(2n)^{2n}} & \text{if } n \text{ is even.} \end{cases}$$

- (a) Show that the Ratio Test is inconclusive for $\sum_{n=1}^{\infty} b_n$.
- (b) Use the Root Test to show that $\sum_{n=1}^{\infty} b_n$ converges.

The next part of this question involves constructing series for which neither the Ratio Test nor the Root Test are applicable.

- (3) Construct series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ such that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$ and $\lim_{n \rightarrow \infty} \frac{b_{n+1}}{b_n}$ do not exist, but $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} b_n$ diverges. [*Hint.* For a_n consider the sequence whose even terms are given by $\frac{1}{n^2}$ and whose odd terms are given by $\frac{1}{(2n+1)^2}$.]
- (4) Construct series $\sum_{n=1}^{\infty} c_n$ and $\sum_{n=1}^{\infty} d_n$ such that $\lim_{n \rightarrow \infty} \sqrt[n]{c_n}$ and $\lim_{n \rightarrow \infty} \sqrt[n]{d_n}$ do not exist, but $\sum_{n=1}^{\infty} c_n$ converges and $\sum_{n=1}^{\infty} d_n$ diverges. [*Hint.* For c_n , try the sequence from the previous hint.]

Finally, after having thought about the Ratio and Root Tests for so long, consider the series below and (using any test!) determine if they converge or diverge.

- (5) (a) $\sum_{n=1}^{\infty} \left(\sum_{k=1}^n \frac{1}{k} \right)^n$ (b) $\sum_{n=1}^{\infty} \frac{\sin\left(\frac{1}{n!}\right)}{\cos\left(\frac{1}{n!}\right)}$ (c) $\sum_{n=2}^{\infty} \left(\frac{n!}{n^n} \right)^n$.

Question 4 – Convergence and Absolute Convergence

- (1) Suppose that $\sum_{n=1}^{\infty} a_n$ converges and is not necessarily positive. Show that $\sum_{n=1}^{\infty} a_n^2$ need not converge.
- (2) Suppose that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. Must $\sum_{n=1}^{\infty} (a_n + a_{n+1})$ be absolutely convergent. Justify your answer.
- (3) Suppose that $\lim_{n \rightarrow \infty} a_n = L \neq 0$ and that $a_n \neq 0$, for all $n \in \mathbb{N}$. Show that $\sum_{n=1}^{\infty} |a_{n+1} - a_n|$ converges if, and only if, $\sum_{n=0}^{\infty} \left| \frac{1}{a_{n+1}} - \frac{1}{a_n} \right|$ converges.

Recall that a convergent series $\sum_{n=1}^{\infty} a_n$ is called **conditionally convergent** if $\sum_{n=1}^{\infty} |a_n|$ diverges. For the remainder of this question, we will examine what rearrangements of conditionally convergent series can look like. For the remainder of this question suppose that $\sum_{n=1}^{\infty} a_n$ is a conditionally convergent series. Without loss of generality we may assume that $a_n \neq 0$, for all $n \in \mathbb{N}$ (**Bonus:** Why?)

- (4) Let $S \in \mathbb{R}$. We will prove that there is a rearrangement $(b_n)_{n \in \mathbb{N}}$ of $(a_n)_{n \in \mathbb{N}}$ such that $\sum_{n=1}^{\infty} b_n = S$.
 - (a) Show that $\sum_{n=1}^{\infty} a_n$ has infinitely many positive and infinitely many negative terms.
 - (b) Consider the following two sequences:

$$a_n^+ = \begin{cases} a_n & \text{if } a_n > 0; \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad a_n^- = \begin{cases} a_n & \text{if } a_n < 0; \\ 0 & \text{otherwise.} \end{cases}$$

From (1) it follows that each subsequence has infinitely many non-zero terms. Show that:

$$\sum_{n=1}^{\infty} a_n^+ \quad \text{and} \quad \sum_{n=1}^{\infty} a_n^-$$

must both diverge.

- (c) Show that there is some $p \in \mathbb{N}$ such that $\sum_{n=1}^p a_n^+ > S$. Let p_1 be the smallest such positive integer.
- (d) Show that there is some $q \in \mathbb{N}$ such that $\sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{q_1} a_n^- < M$. Let q_1 be the smallest such positive integer.

- (e) Briefly explain why for all $n \in \mathbb{N}$ there must be some (smallest) p_n such that:

$$M < \sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{q_1} a_n^- + \sum_{n=p_1}^{p_2} a_n^+ + \sum_{n=q_1}^{q_2} a_n^- + \cdots + \sum_{n=p_{n-1}+1}^{p_n} a_n^+,$$

and some (smallest) $q_n \in \mathbb{N}$ such that:

$$\sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{q_1} a_n^- + \sum_{n=p_1}^{p_2} a_n^+ + \sum_{n=q_1}^{q_2} a_n^- + \cdots + \sum_{n=p_{n-1}+1}^{p_n} a_n^+ + \sum_{n=q_{n-1}+1}^{q_n} a_n^- < M.$$

- (f) Consider the new sequence:

$$a_1^+, \dots, a_{p_1}^-, a_1^-, \dots, a_{q_1}^-, a_{p_1+1}^+, \dots, a_{p_2}^-, a_{q_1+1}^-, \dots, a_{q_1}^-, \dots$$

It is now relatively easy to show that:

$$\sum_{n=1}^{p_{i+1}-1} a_n^+ + \sum_{n=1}^{q_i} a_n^- \leq M < \sum_{n=1}^{p_i+1} a_n^+ + \sum_{n=1}^{q_i} a_n^-,$$

for all $i \in \mathbb{N}$. [**Bonus:** Show this.] Deduce from the above that:

$$0 < \left(\sum_{n=1}^{p_{i+1}} a_n^+ + \sum_{n=1}^{q_i} a_n^- \right) - M \leq a_{p_{i+1}}^+.$$

- (g) Conclude that the $(p_{i+1} + q_i)$ -th partial sum of the new series converges to M .

At this point the proof is almost done. Similarly to (g) the $(p_{i+1} + q_{i+1})$ -th partial sum also converges to M , but then, the whole sequence of partial sums must also converge to M , so (after dropping all the new terms which are zero) we have produced a rearrangement of the series which converges to M .

- (5) **Bonus:** Can you briefly explain how to adapt the argument from (4) to produce a series which diverges to $+\infty$? What about a series that does not converge to any limit?

Part (4) of this question is relatively hard. Any good attempt (even if not fully correct) will be awarded full marks. The goal of this question is to give you some interesting food for thought, not to ruin your Thanksgiving!

Question 5 – Power Series

Recall from Q.6(2) of HW3 that the Fibonacci sequence is given by:

$$F_1 = 1, F_2 = 1, \text{ and } F_{n+1} = F_n + F_{n-1}, \text{ for } n > 2.$$

Recall that in (b) of that question, you had to show that $\frac{F_{n+1}}{F_n} < 2$, for all $n \in \mathbb{N}$.

The goal of this question is to find an explicit formula for the n -th term of the Fibonacci sequence. Let's start by defining the following function:

$$f(x) = \sum_{n=0}^{\infty} F_n x^n = 1 + x + 2x^2 + 3x^3 + 5x^4 + \cdots$$

- (1) Show that the Taylor series for f converges for $|x| < \frac{1}{2}$.
- (2) Let the power series $\sum_{n=0}^{\infty} b_n x^n$ be defined as the Taylor series of the function $f(x)(1 - x - x^2)$. Show that $b_0 = 1$, $b_1 = 0$ and that $b_n = a_n - a_{n-1} - a_{n-2}$, for $n \geq 2$.
- (3) Using the previous part, deduce that:

$$f(x) = \frac{-1}{x^2 + x - 1},$$

for $|x| < \frac{1}{2}$.

- (4) Let the roots of $x^2 + x - 1$ be c_1 and c_2 . Use partial fractions to write:

$$\frac{-1}{x^2 + x - 1} = \frac{A}{x - c_1} + \frac{B}{x - c_2},$$

for appropriate constants A, B .

- (5) Use the previous parts to find a single power series about 0 for $f(x)$.
- (6) Show that $\frac{1}{c_1} = \frac{1+\sqrt{5}}{2}$ and $\frac{1}{c_2} = \frac{1-\sqrt{5}}{2}$. Using the fact that the Taylor series for $f(x)$ is unique, show that:

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right],$$

for all $n \in \mathbb{N}$.